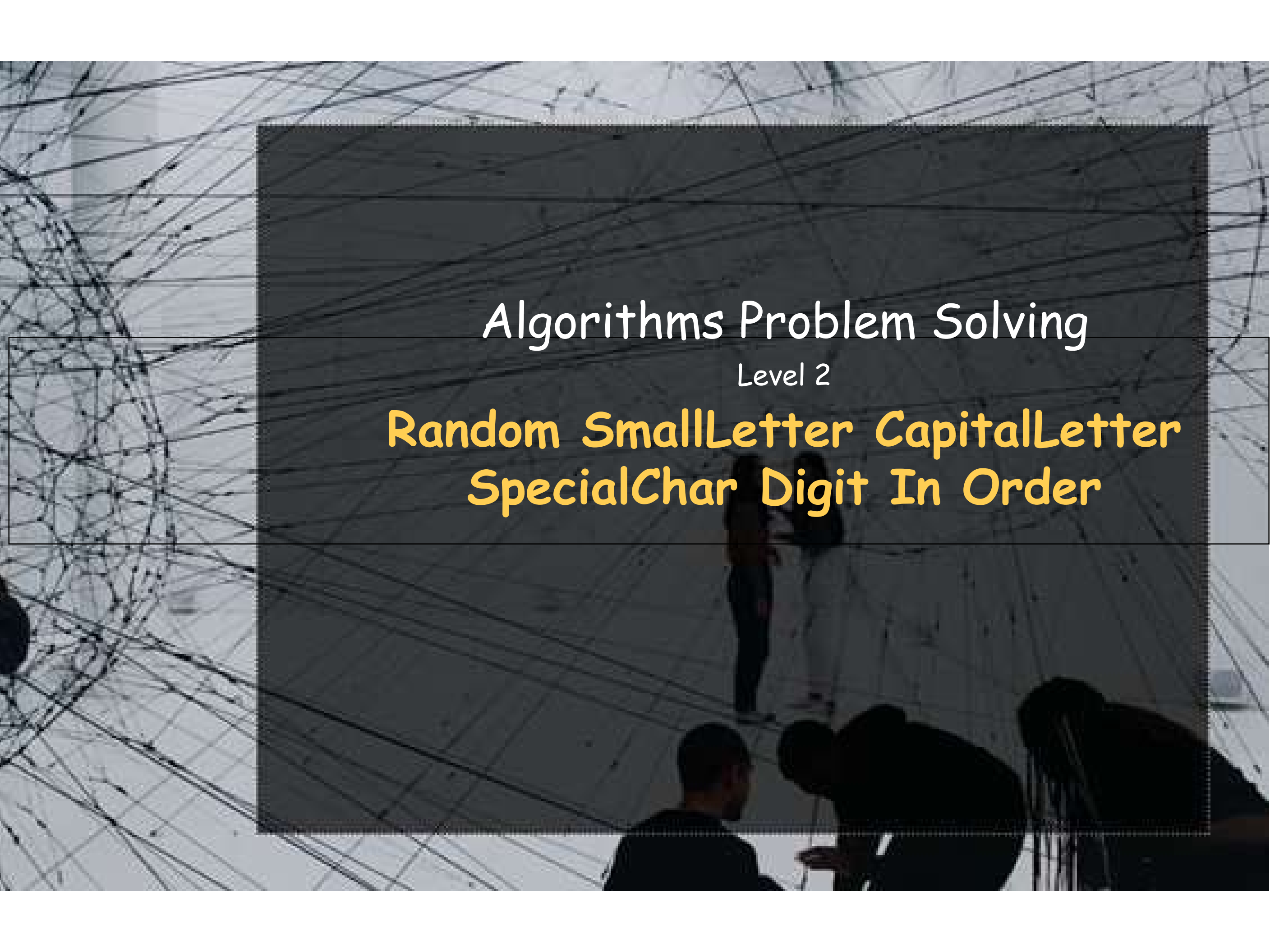


Algorithms Problem Solving

Level 2

Random SmallLetter
CapitalLetter SpecialChar
Digit In Order



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Random SmallLetter CapitalLetter
SpecialChar Digit In Order

Definition

- ◆ A program that generates and prints four random characters in a specific order.
- ◆ The order is:
 - 1 Small letter
 - 2 Capital letter
 - 3 Special character
 - 4 Digit

Explanation

- ◆ The program uses an `enum` called `enCharType` to represent the different character types.
- ◆ `SmallLetter = 1` represents lowercase letters from `a` to `z`.
- ◆ `CapitalLetter = 2` represents uppercase letters from `A` to `Z`.
- ◆ `SpecialCharacter = 3` represents special characters in the selected ASCII range.
- ◆ `Digit = 4` represents digits from `0` to `9`.
- ◆ `RandomNumber()` generates a random number between two given values.
- ◆ `GetRandomCharacter()` receives the required character type and uses a `switch` statement to generate a character from the correct range.
- ◆ The `main()` function calls `GetRandomCharacter()` four times in the required order.

☐ Steps in main()

- 1☐ Initialize the random number generator using `srand()`.
- 2☐ Request a random `SmallLetter`.
- 3☐ Request a random `CapitalLetter`.
- 4☐ Request a random `SpecialCharacter`.
- 5☐ Request a random `Digit`.
- 6☐ `GetRandomCharacter()` checks the character type using `switch`.
- 7☐ `RandomNumber()` generates a value inside the required character range.
- 8☐ Convert the generated value to `char`.
- 9☐ Print the generated character.

💡 Example

◆ One possible output is:

i
G
&
7

◆ Here:

`i` → Small letter

`G` → Capital letter

`&` → Special character

`7` → Digit

➡ Result

◆ The program prints exactly four random characters in this order:

Small letter

Capital letter

Special character

Digit

◆ The exact characters change each time the program runs.

✈ Note

- ◆ The ``enum`` makes the character types easier to understand and use.
- ◆ ``GetRandomCharacter(enCharType charType)`` uses the selected type to determine which range should be generated.
- ◆ For lowercase letters:
``RandomNumber('a', 'z')``
- ◆ For uppercase letters:
``RandomNumber('A', 'Z')``
- ◆ For digits:
``RandomNumber('0', '9')``
- ◆ For special characters, the program uses:
``RandomNumber(33, 47)``
- ◆ Character values can be used directly with ``RandomNumber()`` because characters have numeric character codes.

⊘ Warning

- ◆ The special-character range ``33`` to ``47`` does not represent every possible special character. It represents characters in that part of the character-code table, including symbols such as ``!``, ``"`,` ``#``, ``$``, ``%``, ``&``, ``'``, ``(``, ``)``, ``*``, ``+``, ```,``, ``-``, ``.``, and ``/``.
- ◆ The ``break`` statements after ``return`` are unreachable because ``return`` already exits the function. They can be removed.
- ◆ The code uses ``rand()``, ``srand()``, and ``time()`` but does not explicitly include ``<cstdlib>`` and ``<ctime>``. It is better practice to include the required headers.

Comparison

- ▣ `SmallLetter` → generates `a` to `z`.
- ▣ `CapitalLetter` → generates `A` to `Z`.
- ▣ `SpecialCharacter` → generates characters with codes `33` to `47`.
- ▣ `Digit` → generates `0` to `9`.
- ▣ `enum` → gives meaningful names to the different character types.
- ▣ `switch` → selects the correct character range based on the requested type.

✦ Summary

- ✓ ``enum`` is used to represent different character types.
- ✓ ``RandomNumber()`` generates a random value inside a specified range.
- ✓ ``GetRandomCharacter()`` generates a character based on its type.
- ✓ Lowercase letters use ``a`` to ``z``.
- ✓ Uppercase letters use ``A`` to ``Z``.
- ✓ Special characters use character codes ``33`` to ``47``.
- ✓ Digits use ``0`` to ``9``.
- ✓ The program prints the characters in the order: Small letter → Capital letter → Special character → Digit.

Thank you 😊